

Shantanu Shastri

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EDUCATION

Carnegie Mellon University December 2026
Master of Information Systems Management, GPA: 3.85

Manipal Institute of Technology August 2022
Bachelor of Electronics and Communications Engineering with Minor in Data Science, GPA: 3.99

Relevant Coursework: Data Structure & Algorithms, Java, Python, Object Oriented Programming, Machine Learning, Linear Algebra

SKILLS

Languages & Libraries: Java, C++, JavaScript, Python, TypeScript, SQL, NumPy, Pandas, Bash/Shell, PyTorch
Developer Tools/Frameworks: Spring Boot, Apache, gRPC API, System Design, Node.js, SQL Databases, SonarQube, Veracode, Git, JUnit, Postman, Angular, TensorFlow, Deep Learning, NLP, Linux, Grafana, Redux, Flask, React.js, FastAPI, HTML/CSS
Infra & Database: AWS, GCP, Azure, Docker, Kubernetes, Terraform, Kafka, MySQL, MongoDB, Redis, Neo4j, Azure Blob, GraphQL

EXPERIENCE

BlackRock Gurgaon, India
AI Engineer Intern June 2026 – Present

- Multi-Agent Systems & Event Sourcing: Building a Multi-Agent system to handle post-trade operations, creating a semantic layer using Protobuf and event sourcing via Apache Kafka.
- Hierarchical RAG & Agent Learning: Developing a hierarchical multi-agent system running on Knowledge Graph RAG that creates patterns to identify and learn from user actions.
- Evaluation & Policy Optimization: Created offline and online evaluation frameworks tracking latency, confidence scores, and data drift, utilizing policy optimization to improve agent accuracy for a system handling 3.5M records and \$14T in cash.

Software Developer 2 - Associate January 2025 – July 2025
Trade routings remodeling - Post Trade

- System Scalability & Optimization: Architected a central data framework and TTL Azure Blob microservices to resolve legacy bottlenecks, successfully automating 1.5M+ touchpoints across 117 clients while significantly reducing infrastructure storage costs.
- gRPC API & Caching: Engineered a distributed read-through Ignite cache to address high-latency APIs, cutting response times by 3x.

Software Developer 1 - Analyst July 2022 – December 2024
Trade routings remodeling - Post Trade

- Database Optimization: Achieved 37% reduction in query latency by refactoring critical legacy database tables and leveraging composite indexes for faster data retrieval, significantly improving user experiences for key workflows.
- Full-stack Development: Spearheaded end-to-end development of responsive, high-performance Angular-based UI using NgRx redux state management and integrated Azure's LLM APIs to deliver real-time AI capabilities directly into the user workflow.

Multi-Instance – Multi-Threading

- Distributed Systems Architecture: Resolved critical singleton bottlenecks by engineering a sharded, multi-instance server environment with Zookeeper leader election, effectively distributing load and boosting system performance by 24%.
- DevOps & Quality Engineering: Built comprehensive CI/CD pipelines using Azure DevOps, with integrated JUnit/Jacoco testing suites, reducing time-to-market by 40% while maintaining SONAR scores >90%.

PROJECTS

High-Performance Modular Booking Architecture

December 2025

- Backend Scalability: Engineered a decoupled microservices architecture using Apache Kafka, successfully handling bursts of 100,000+ real-time requests with minimal latency spikes.
- Infrastructure & DevOps: Orchestrated containerized deployments using Docker to ensure environment consistency and enable seamless horizontal scaling across production systems.

Zulip Open-Source LLM Extensions

January 2026

- LLM Feature Engineering: Extended the Zulip platform by building a “Message Recap” engine using Tornado, Redis and RabbitMQ that leverages Large Language Models (LLMs) to synthesize unread streams into concise, clickable summaries.
- Cost & System Optimization: Designed a cost-efficient Topic Drift Detection system to correct conversation titles in real-time, optimizing interaction models to significantly minimize token usage costs.

Real-Time Generative AI Product Agent

October 2025

- Generative AI & Vector Search: Developed an AI agent utilizing Meta’s FAISS for high-speed vector retrieval to generate high-fidelity product images from text descriptions.

Production Recommendation & MLOps System

May 2026

- Hybrid Recommender & Serving: Developed a hybrid recommender using LightFM (WARP) and SVD++ (722 recs/s throughput) with Llama-3-based cold-start logic, containerizing the service with Docker and a deterministic SHA-256 load balancer.
- Automated MLOps & Auditing: Built a 4-hour retraining loop with zero-downtime model reloading tracked via MLflow, monitored metrics using Prometheus/Grafana (61.4% online hit rate), and performed STRIDE, STPA, and demographic fairness audits.